MATH 126 C
Exam II
Winter 2019

Name _____ Student ID #_____

HONOR STATEMENT

"I affirm that my work upholds the highest standards of honesty and academic integrity at the University of Washington, and that I have neither given nor received any unauthorized assistance on this exam."

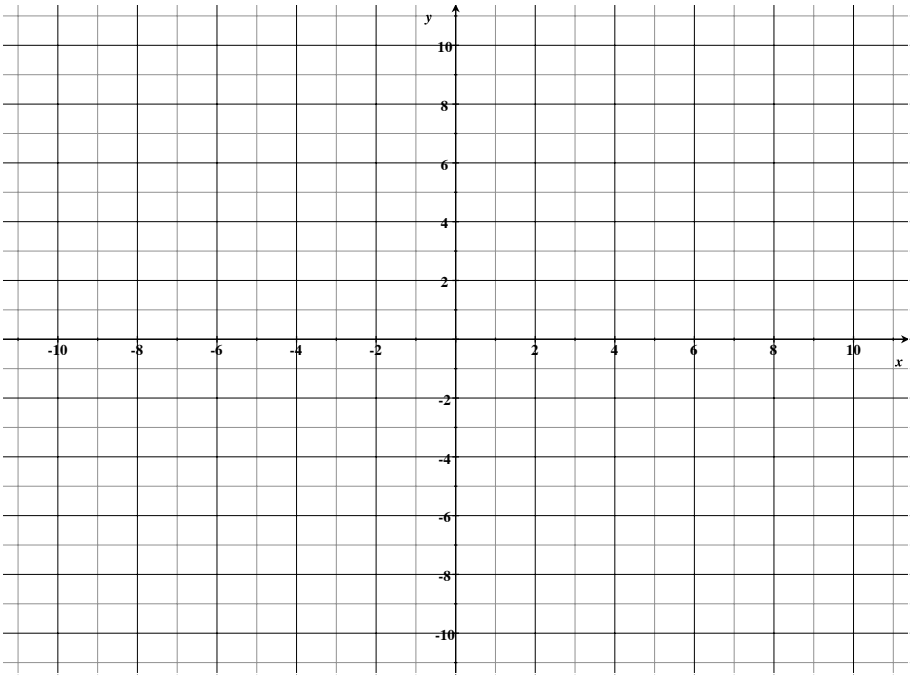
SIGNATURE:_____

- Your exam should consist of this cover sheet, followed by 5 problems. Check that you have a complete exam.
- Pace yourself. You have 50 minutes to complete the exam and there are 5 pages. Try not to spend more than about 10 minutes on each page.
- Unless otherwise indicated, **show all your work and justify your answers**.
- Unless otherwise indicated, your answers should be exact values rather than decimal approximations. (For example, $\frac{\pi}{4}$ is an exact answer and is preferable to its decimal approximation 0.7854.)
- You may use a **TI 30XII S** calculator and one 8.5×11-inch sheet of handwritten notes. **All other calculators, electronic devices, and sources are forbidden**.
- **Do not write within one centimeter of the edge of the page.**
- If you need more room, use the back of this cover sheet and the back of the last page and **tell your grader where to look for your solution**. If you need more room than that, ask your TA for extra paper and put your name on it.
- The use of headphones or earbuds during the exam is not permitted.
- There are multiple versions of the exam, you have signed an honor statement, and cheating is a hassle for everyone involved. DO NOT CHEAT.
- You are not allowed to use your phone for any reason during this exam. **Turn your phone off and put it away for the duration of the exam**.

GOOD LUCK!

USE THIS PAGE IF YOU NEED MORE ROOM.
TELL YOUR GRADER WHERE TO FIND YOUR WORK.

1. (8 points) Find, sketch, and shade the domain of $f(x, y) = \sqrt{100 - x^2 - y^2} + \ln(x - y)$.



2. (10 points) Use implicit differentiation to compute $\dfrac{\partial z}{\partial x}$ and $\dfrac{\partial z}{\partial y}$:

$$\sin(x^2 z) = 8ye^z.$$

3. (12 points) Find and classify all critical points of the function

$$f(x, y) = 3x - x^3 - 6xy^2.$$

4. (10 points) Evaluate

$$\int_1^2 \int_1^{x^2} \left(xe^y + \frac{1}{x^3} \right) dy\, dx.$$

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Chapter 15 material - covered in Math 224

5. (10 points) Let V be the volume of the solid that lies below the surface $z = xy$ and above the triangle in the xy-plane with vertices $(0, 0)$, $(0, 3)$, and $(3, 1)$.

YOU DO NOT NEED TO COMPUTE THE VOLUME.

Fill in the boxes below to give the iterated integrals one would need to evaluate in order to find the volume V. Show your work.

$$V = \int_{\square}^{\square} \int_{\square}^{\square} \square \; d\square \, d\square$$

YOU DO NOT NEED TO COMPUTE THE VOLUME.
JUST FILL IN THE BOXES.

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